1 Basic AIP Display

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In general, *AIP* has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons within the touch screen or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these selection lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.

To ensure proper processing and posting, terminals with "MDE" operation mode must <u>not</u> be switched off during times without shift.

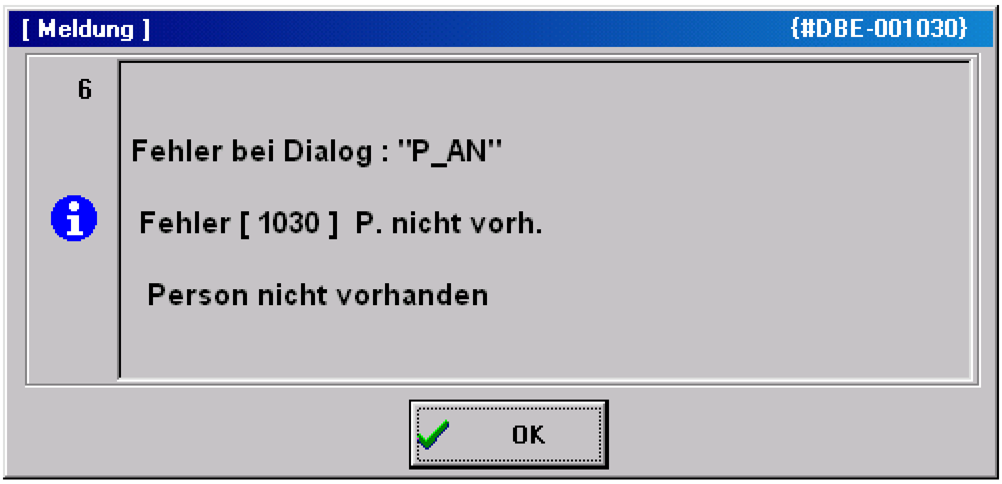
1.1 Basic displays – header and footer

Header

The *AIP* logo is displayed top left of the screen, which may be exchanged by a customer logo after corresponding configuration.

Possible messages (e.g. if a dialog is opened for more than five minutes) are displayed to the right of it.

A separate window opens to display error messages that occur during data collection (e.g. validity checks).



Basic displays

A maximum of 16 workplaces or machines can be assigned to the *AIP* terminal. The individual workplaces can be found within the list area in the order in which they have been assigned to the terminal in MOC.

As regards the basic display of the *AIP* terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured within the terminal configuration at the client. The individual basic views are described in the sections that follow.

Footer

The MPDV logo can be found at the bottom left of the screen. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

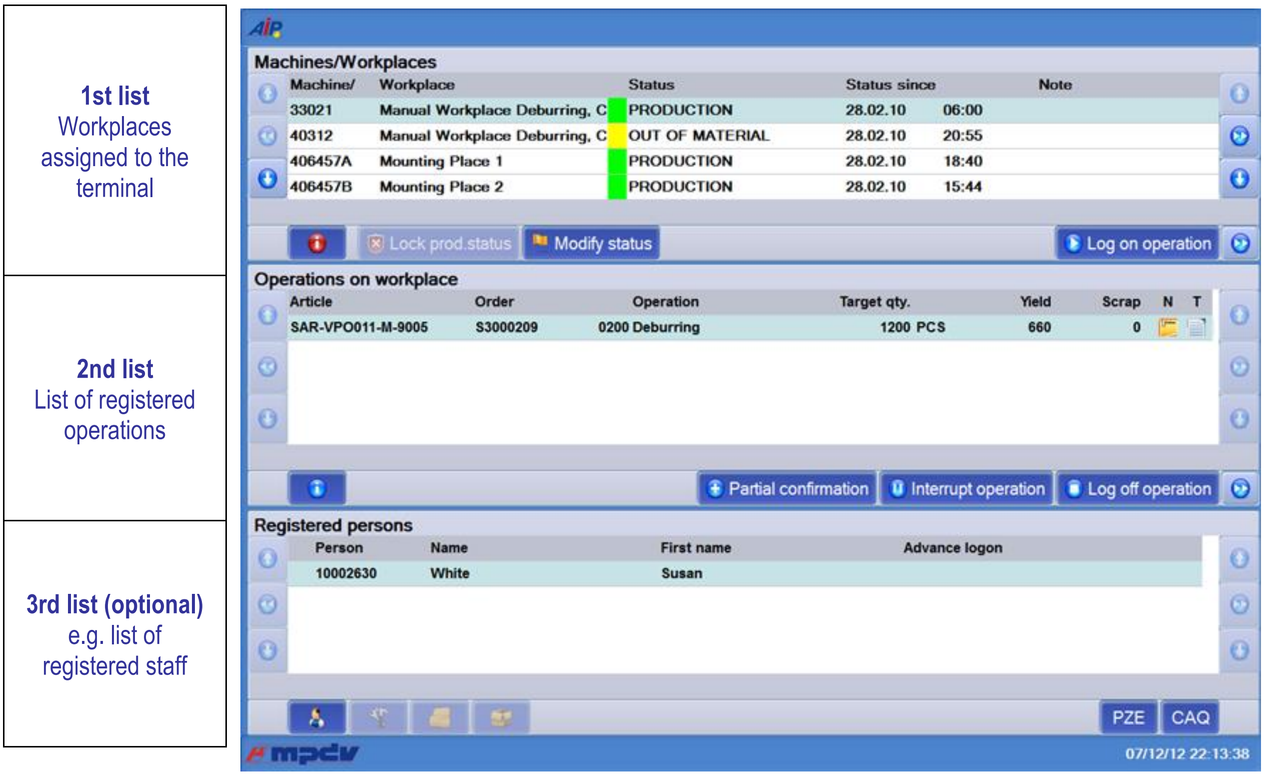
Further information is displayed in the center : the current terminal status, AIP version number, date of the build, IP address of the server as well as the terminal number.

The current date and time are displayed to the right.

AIP status

	Network connection has been established The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	Commands are being transferred to the server
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.
	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

1.2 Basic display “tabular view“



The tabular basic display consists of two or three tables, subject to the configurations made. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional display).

“Machines/workplaces" table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as the designation are displayed.

Status

The status is displayed in color and as status text. Coloring is as follows:

- green: Production
- yellow: Assigned status
- red: Not assigned

Status since

Point in time since the status is available.

For ADE workplaces[1] the point in time refers to the last manual status change, for MDE workplaces it refers to the time when the last status change was identified (for machine connections), to the point in time when the status was changed manually the last time or to the time of the last shift change.

Please note

It is indicated here if the “block production status” function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.

Please note

By way of “customizing” services it is possible to adapt the layout of display lists, displayed data fields, sort sequences, etc. according to the customer’s requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic display of terminals. The software does not allow it.

Provided that the "compensate manual quantities" option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately, once they have been entered but only once lists have been reloaded.

"Operations at workplace" table

The second table shows the operations that are currently logged on to the selected workplace. The following columns are displayed:

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number.*

Target quantity

Target quantity that is defined for the operation.

[1] We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the “MDE” operation mode. In any other case, it is an ADE workplace.

Yield

Yield which has already been produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity which has already been produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note, which should be visible at the terminal, has been recorded for this operation in the graphic planning board at the client. The note(s) is/are displayed by clicking the OP info button ( ).

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button).

Below the second list there is a line that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured respectively. Which information is displayed in this list depends, among other things, on the workplace configuration.

The following lists can be displayed:

- Staff logged on to the currently selected workplace (BDE)
- Resources logged on to the currently selected workplace (WRM)
- Materials/input batches logged on to the currently selected workplace (MPL/TRT)
- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow for switching between these lists.

Please note

The registered staff displayed in the third list correspond to the list of the dialog “F5 registered persons…”. Selecting a person in the third list does *not* affect selection of the operation in the list of OPs running at the workplace and, as a result, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic display

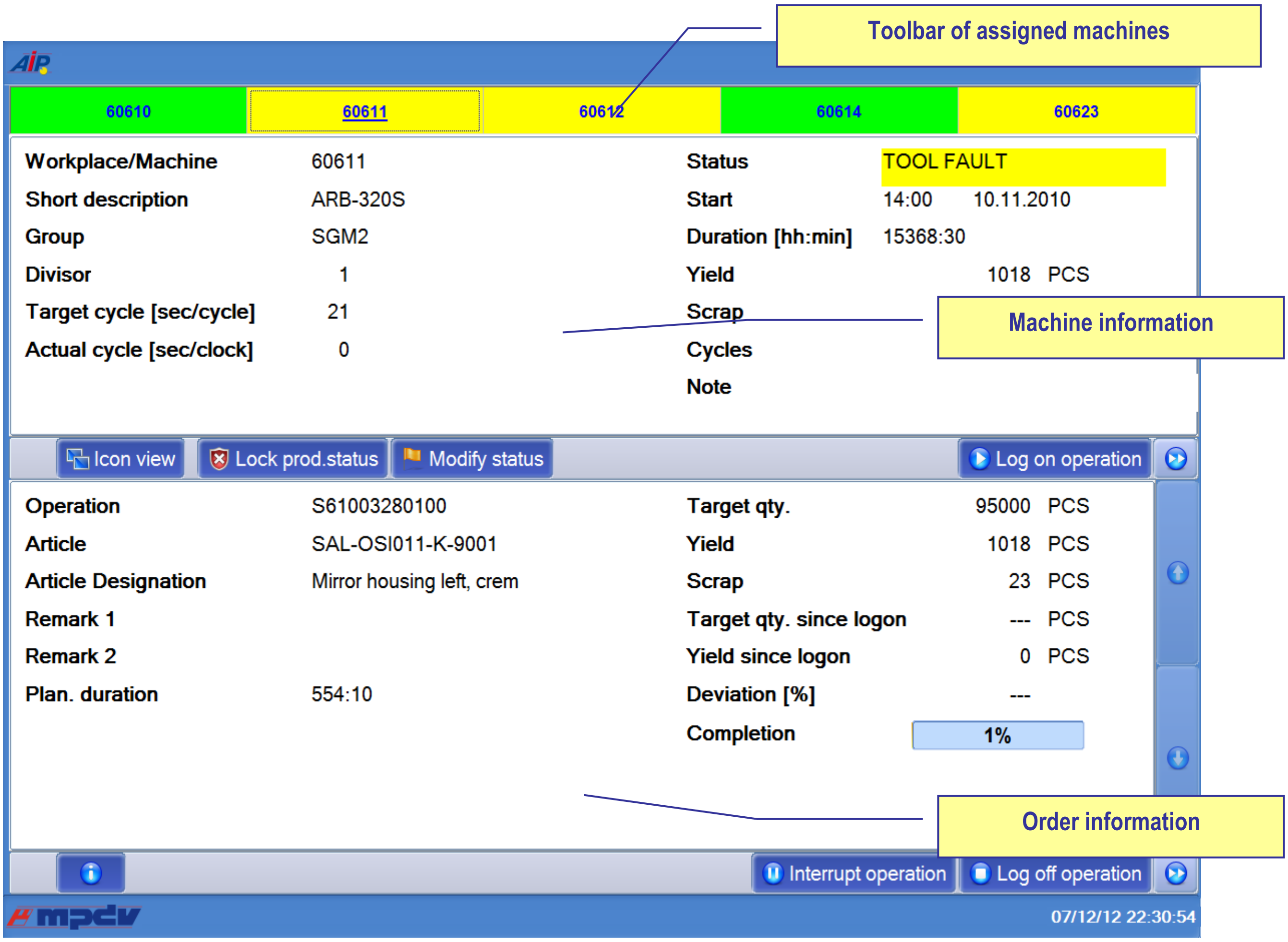
A toolbar, which may be configured by customizing, is assigned to each list in the basic display. This makes the purpose of the function clear to the user. The “partial upload/confirmation” function can be found, for example, below the list of registered operations.

In fact, the toolbar may include several “tabs”, which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. change partitioning) can be opened by clicking the corresponding button.

Please note
The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

1.3 "Machine overview" basic display

If the “change view” button is clicked in the basic display, the view changes to the following presentation:



This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation altogether has three areas:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:
- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines as well as to shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. Provided that several orders/OPs are logged on, it can be switched between them using arrow keys that are then shown additionally.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$Displayed\ \ partitioning = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + ... \right]$$

TLG_M	Partitioning of the machine (TLG in mnr.lst)
DIV_M	Pulse factor of the machine (IMPFAKT in mnr.lst)
TLG_{AGi}	Partitioning of the individual order (TLG in anr.lst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview at the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal may get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) at the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time in which the production lock of the machine has not been active). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.
Calculation:
TargetSinceLogon = NetRunningTime[sec] * Partitioning/TargetCycle[sec/stroke]
NetRunningTime: Time since logon, in which the production lock has not been set. This calculation does not take into account any breaks that might be defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

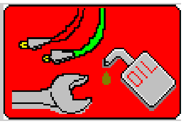
Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced “since logon”.
Calculation: Deviation[%] = 100% * (YieldSinceLogon - TargetSinceLogon) / TargetSinceLogon

Completion

The bar represents the proportion of “yield”, which has been produced until now, compared to the “target quantity”.

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced by a picture showing a yellow or red oilcan, depending on the maintenance activity that is required:



1.4 “Machines as icons” basic display

This view can be configured as the default display within the terminal configuration at the client. It has the advantage that the user can tell from a distance whether or not all machines are in the “Production” status.

All MDE machines have their own buttons colored according to the corresponding status:

- green: Production
- yellow: Assigned status
- red: Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

If a button is touched, the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon presentation of machines”.